Nikhil Yadav

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PROFILE

Computer Science graduate with interests in cybersecurity, cryptography, network security, and hands-on security testing in controlled lab environments. Experienced with security tools, digital-forensics workflows, and technical event coordination.

TECHNICAL SKILLS

Languages: C, C++, Python, JavaScript, HTML/CSS, SQL, scripting

Tools: Git/GitHub, Linux Shell, VS Code, Metasploitable, Burp Suite, Nmap, Metasploit

Concepts: CTFs, cryptography basics, networking fundamentals, technical security awareness, password-cracking concepts

PROJECTS

Enhanced Lightweight Hashing for Low-Power Devices | C, C++, Bash · Mar 2024

• Explored optimizations to PHOTON and BLAKE lightweight hashing algorithms for IoT devices and constrained environments.

• Focused on cryptographic performance and security properties, including avalanche effect, collision resistance, and entropy.

Predictive Modeling for Machine Failure Detection | Python, HTML, CSS, JavaScript · Oct 2025

• Designed a machine-failure prediction dashboard using interactive web technologies.

• Integrated a Random Forest model to analyze operational data and forecast potential failures.

Developed Vulnerable Machine for Penetration Testing | Nmap, Burp Suite, Metasploit, DVWA · Nov 2024

• Configured intentionally vulnerable lab environments using Metasploitable, DVWA, and OWASP BWA.

• Practiced authorized penetration testing in isolated labs and documented test scenarios.

EXPERIENCE

Core Committee Member — Cyber Sentinel | Cybersecurity awareness & technical event management

• Coordinated cybersecurity events, technical speaker engagement, and event logistics.

• Organized workshops and CTF activities and collaborated with technical teams to promote cybersecurity awareness.

Intern — IBM | Digital forensics & file recovery

• Used TestDisk, PhotoRec, Recuva, and Extundelete for recovery workflows across Windows and Linux environments.

• Analyzed FAT, NTFS, and EXT4 file systems and documented methodologies, outcomes, and tool configurations.

EDUCATION

B.Tech, Computer Science — UPES, Dehradun | 2022–2026 | CGPA: 7.6/10

Senior Secondary — Gurukul B. L. Mohta | 2020–2021 | 88%

Secondary School — R.S.V | 2018–2019 | 92.6%

CERTIFICATIONS & AWARDS

Introduction to Cybersecurity — Cisco-verified course certificate | Nov 2026

AISSE Merit Recognition — Resume lists recognition among the top 0.1% of candidates | Aug 2021